1. Incident Timeline

1.1 Sequence of Events

Time	Event
2300 hrs	Routing to Logistics sector (Charlie) lost due to misconfiguration
2305 hrs	Nmap SYN scan launched from unknown host targeting internal network
2310 hrs	Telnet credentials captured via plaintext transmission
2315 hrs	FTP credentials stolen; unauthorized file transfer observed
2320 hrs	Laptop detected beaconing to suspicious port 4444

2. Evidence of Credential Theft

2.1 FTP Credential Exposure

Wireshark capture revealed plaintext FTP authentication:

Username	gftp
Password	[REDACTED]
Source IP	192.168.1.34
Destination IP	192.168.1.39
Filter Used	ftp

2.2 Telnet Credential Exposure

Telnet session captured showing character-by-character plaintext transmission. Filter used: *telnet*

2.3 DVWA HTTP POST Capture

HTTP POST request captured containing login credentials in plaintext. Filter used: *http.request.method == POST*

3. Attacker Identification

Indicator	Value
Attacker IP	192.168.1.34 (compromised workstation)
Target IP	192.168.1.39 (internal server)
Attack Vector	Network sniffing of unencrypted protocols
Tools Used	Nmap, Wireshark (or equivalent)

4. Defensive Controls That Failed

Missing Control	Impact

No encryption on FTP	Credentials transmitted in plaintext
No encryption on Telnet	Every keystroke visible to attackers
No HTTPS on web apps	HTTP POST data exposed
No active IDS monitoring	Reconnaissance scans went undetected
Routing misconfiguration	Network segmentation failure

5. 10-Minute Containment Plan

Immediate actions to stop the attack:

1. Isolate compromised laptop from network (unplug cable or disable interface)
2. Block attacker IP at perimeter firewall
3. Disable compromised user accounts immediately
4. Stop FTP and Telnet services: systemctl stop vsftpd inetd
5. Capture forensic image of affected systems
6. Alert security team and begin log preservation

6. Long-Term Remediation Plan

6.1 Protocol Replacements

Insecure	Secure	Reason
Telnet (23)	SSH (22)	Encrypted tunnel protects all data
FTP (21)	SFTP (22)	File transfer over encrypted SSH
HTTP (80)	HTTPS (443)	TLS encryption for web traffic

6.2 Monitoring Improvements

1. Deploy Suricata IDS with custom rules for reconnaissance and credential exposure
2. Implement SIEM for centralized log collection and correlation
3. Enable NetFlow monitoring on all routers
4. Configure alerts for unusual outbound connections (e.g., port 4444)
5. Establish baseline network behavior for anomaly detection
6. Regular vulnerability scanning and penetration testing